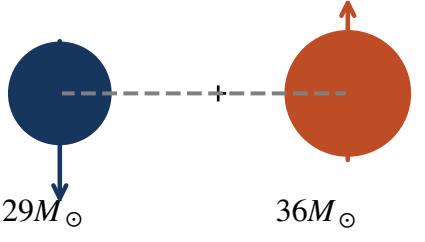
$$l^3 = GM/(2\pi f)^2$$

题设半径 $r_i = Gm_i/c^2$